The Meaning and Grammar of Pure Gestures: Theoretical Insights*

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June 10, 2025

Abstract. While studies of gestures generally focus on their co-speech use, gestures can also (more rarely) be produced on their own, with a dedicated time slot. The meaning and grammar of such 'pure' gestures are easier to analyze because they don't co-occur with words. Pure gestures have led to three key findings in recent theoretical linguistics. First, the informational content of an iconic gesture can be divided 'on the fly' among traditional slots of the inferential typology of language, which includes assertions, presuppositions, implicatures, etc. This finding extends to novel gestures, visual animations and (gradient and iconic) classifier predicates in sign language, which suggests that the inferential typology is mostly derived through productive rather than lexical means. Second, some pure gestures (such as pointing and repetition-based plurals) have a grammar reminiscent of some constructions of sign languages (despite the fact that the latter are not gestural systems). While this could suggest that, within the visual modality, Universal Grammar specifies certain form-to-function mappings, an alternative is that the realization of these elements has deeper cognitive roots. Finally, pure gestures have been found to have a dedicated 'iconic syntax', with a preference for preverbal or postverbal objects depending on whether the object denotation is typically visualized before or after the relevant action; here too, this result dovetails with findings from sign language classifier predicates. Overall, these results highlight the importance of gestural studies for theoretical linguistics, and the fruitfulness of multimodal investigations that include both gestures and sign languages.

Keywords: gestures, pure gestures, pro-speech gestures, post-speech gestures, co-speech gestures, inferential typology, classifier predicates, gestural grammar, iconic syntax

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***Funding:** This research received funding from the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation programme (grant agreement No 788077, Orisem, PI: Schlenker). Research was conducted at DEC, Ecole Normale Supérieure - PSL Research University. DEC is supported by grant FrontCog ANR-17-EURE-0017.
Acknowledgments: We are grateful to Lucie Ravaux for help with the manuscript and bibliography, as well as to Emmanuel Chemla, Léo Migotti and Lyn Tieu and Emmanuel Chemla.
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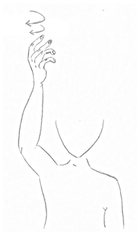
1 Introduction

1.1 Goals

While most uses and studies of gestures involve co-speech gestures, which co-occur with words, this piece is devoted to gestures that occur on their own, with a dedicated time slot—what may be called 'pure' gestures (they are also called 'pro-speech' and 'post-speech' gestures, depending on whether they fully replace words with a grammatical function, or follow words they modify).[1] Pure gestures are certainly far less common than co-speech gestures, but their study has made three important contributions to recent theoretical linguistics[2] (we only focus on the gestures of hearing speakers; for a psycholinguistic approach to the gestures of Deaf signers, see for instance Emmorey 1999; for a psycholinguistic approach to homesigns—used by deaf individuals without access to a sign language, see for instance Goldin-Meadow 2024).

The first finding is that the informational content of an iconic gesture can be divided 'on the fly' among traditional slots of the inferential typology of language, which includes assertions, presuppositions, implicatures, etc. This finding extends to novel gestures, visual animations and certain constructions of sign language, called classifier predicates, whose position and movement is free and gradient, and interpreted iconically (e.g. Emmorey and Herzig 2003, Zwitserlood 2012, Zucchi 2017). This suggests that the inferential typology is, for the most part, derived through productive rules rather than encoded in arbitrary lexical entries. To illustrate, consider example (1), where the verb is replaced with a gesture of unscrewing a bulb from a high position (e.g. from the ceiling). Even if one hasn't seen this gesture before, it conveys information about the position of the bulb. Crucially, this information is treated as a presupposition, as shown by the fact that the 'high position' inference is preserved in the question.

(1) The light bulb in that room, are you going to  ?
(Schlenker 2022; illustration: Marion Bonnet[3])

The second finding is that some pure gestures, such as pointing and plurals, have functional roles reminiscent of some sign language constructions (despite the fact that sign languages are of course not gestural systems[4]). While this could suggest that in the visual modality 'Universal Grammar' does not just specify the structure but also the form of some grammatical elements, an alternative is that the realization of these elements has deeper cognitive sources. To illustrate, consider the example in (2):

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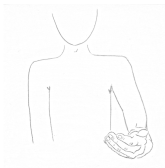
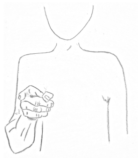
[1] For a recent survey of co-speech gestures, which are not considered in this piece, we recommend Ebert 2024. For a general introduction to gestures, see for instance Kendon 1997, 2004, McNeill 1992, 2005, Abner et al. 2015 and Goldin-Meadow 2024, as well as a Ladewig 2020 for a cognitive grammar approach. On a terminological level, when talking about sequences of pure gestures, the term 'silent gestures' is often used (e.g. in Goldin-Meadow et al. 2008, Goldin-Meadow 2024). However pro-speech and post-speech gestures are often naturally accompanied with onomatopoeias, and thus it would not be judicious to call them 'silent gestures'. As mentioned by Fritzsche (to appear),"pro-speech gestures are also referred to as *speech-framed* or *speech-linked* (McNeill, 2005), or gestures that replace speech in *interrupted utterances* (Ladewig, 2020)." Clark 2016 uses the term 'embedded depiction' for a category that includes pro-speech gestures, while Hsu et al. 2021 speak of 'speech-embedded non-verbal depictions'.

[2] This survey borrows from several pieces we wrote on pure gestures and other iconic constructions, in particular Schlenker 2019, 2020, 2022, Schlenker et al. 2024, and Schlenker et al., to appear.

[3] Illustrations by Marion Bonnet are reprinted under Creative Commons CC BY 4.0.

[4] For sign languages, see for instance Sandler and Lillo-Martin 2006 and Emmorey 2023.

(2)   

-right  -left
Whenever I can hire [a mathematician] or [a sociologist], I pick -right.
(Video: https://youtu.be/nU5xdXLV43c 1st sentence; Schlenker 2020, 2022; illustrations: Marion Bonnet)

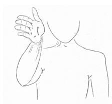
The first disjunct *a mathematician* is pronounced with an open hand on the right while the second disjunct *a sociologist* co-occurs with an open hand on the left. We will argue that these are gestural counterparts of 'loci', positions in signing space that instantiate discourse referents or variables. As a result, when the sentence-final object of *pick* is replaced with a pointing gesture towards the right, we obtain a sentence that is acceptable, and has a reading on which the gestural 'pronoun' is dependent on the indefinite *a mathematician*.

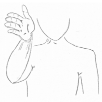
Third, pure gestures have been found to have a dedicated 'iconic syntax', with a preference for preverbal or postverbal objects depending on whether the object denotation is typically visualized before or after the relevant action; and this result too dovetails with findings from sign language classifier predicates, the most iconic component of sign language. To be concrete, let us consider speakers of English and French, which have an SVO word order, as in *the crocodile ate up a ball* and *the crocodile spit out a ball*. Two striking facts arise when related contents are produced with sequences of gestures, specifically a gesture representing the crocodile, another one representing the ball, and a third one representing the action of eating. First, the eat-up case favors preverbal objects (e.g. SOV), thus going against the basic word order of those languages. Second, in the spit-out case, SVO order is regained. A possible explanation is that these are instances of a completely different way of combining expressions than is standard in linguistics: iconic gestures appear in the same order as pictures in a comic, with the ball visible before the action in the eat-up case and after the action in the spit-out case, hence the term 'iconic syntax'.

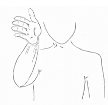
1.2 Types of gestures

Before we delve into the details, we should situate the target gestures within a broader typology. McNeill 2005 distinguishes among four types of gestures: *iconic*, *metaphoric*, *deictic* and *beat*. In a nutshell, iconic gestures resemble their denotations, metaphoric gestures may present "images of the abstract", deictic gestures involve pointing, and beats take "the form of the hand beating time". Several authors (e.g. McNeil 1992, Giorgolo 2010) have an additional subcategory of 'emblems', which are conventionalized gestures with a fixed meaning (e.g. the 'thumb up'  gesture). We will primarily focus on iconic and deictic gestures, but we will take the latter to be part of a broader class of functional gestures, which play roles reminiscent of some grammatical expressions of sign language.

Gestures can also be categorized on the basis of their timing relative to the rest of the sentences they appear in. Some generalizations have been proposed in the literature to connect gesture timing to gesture contribution, on the basis of the typology illustrated in example (3) (see Ebert 2024 for a survey).

(3) a. **Pro-speech gesture:** His enemy, Asterix will _<phhh>.

b. **Co-speech gestures:** Asterix will  **punish** his enemey (the gesture co-occurs with *punish*).

c. **Post-speech gesture:** Asterix will punish his enemy — _<phhh>.
(Illustrations: Marion Bonnet)

First, pro-speech gestures, which fully replace a word with a grammatical function (as in example (3)a), must usually make an assertive contribution (they may make further contributions in addition, e.g. presuppositions or implicatures). Second, while there is general agreement that co-speech gestures (as in example (3)b) are non-assertive, theorists differ as to their nature: some believe that they display the behavior of appositive relative clauses in contributing supplements (Ebert and Ebert 2014), while others take them to trigger presuppositions of a particular sort (Schlenker 2015, 2018). Proponents of the latter analysis have argued that *post*-speech gestures, which follow words they modify (as in example (3)c), display the behavior of appositive relative clauses, whereas others (Barnes and Ebert 2023) take them to make gradient at-issue contributions (see also Ebert 2024). (It should be noted that pro-speech and post-speech gestures are often more natural if accompanied with onomatopoeias, whose content may be minimal; this might be because silence is a bit awkward in a spoken utterance.)

Pro-speech gestures fulfill the grammatical and semantic role of full-fledged words; for this reason, they often lead to ungrammaticality if they are omitted (for further work on pro-speech gestures, see for instance Slama-Cazacu 1976, Clark 1996, 2016, Fricke 2008, Ladewig 2011, 2020, Schlenker 2019, Tieu et al. 2019; see also Jouitteau 2004, Ortega-Santos 2016, and Ebert 2018).[5] They will be the main character in what follows, where we will study how their informational contribution is divided among traditional slots of the inferential typology, how they may realize certain grammatical functions in ways that are reminiscent of some sign language expressions, and how they involve a very different syntax from normal words.

1.3 Semantics and discourse integration of iconic gestures

The first question to ask is how the basic informational content of iconic gestures is derived. Alas, there is no consensus yet on a semantics for the three-dimensional dynamic representations involved in gestures. Recent formal semantics has made heavy use of a highly explicit formal semantics for pictures, developed with great clarity by Greenberg 2013, 2014, 2021, and extended to sequences of silent pictures by Abusch (e.g. 2012, 2020). Pictorial semantics offers an inspiring model because it is particularly transparent: the truth conditions of pictorial representations are derived from the way geometric projections work. In a nutshell, a picture P is true in a situation w relative to a viewpoint v just in case w projects to P relative to v (in view of the details of the geometric projection under study, for instance perspective projection). This framework has been modified to provide an account of the semantics of sign language classifier predicates (Schlenker and Lamberton 2024), but it has not been extended to gestures, which are less constrained. On the other hand, there are recent proposals to account for the semantics of iconic gestures in very different terms; see for instance Giorgiolo 2010 and Lücking et al. 2024.

Another point we leave aside is the discourse integration of gestures. Historically, some of the first formal studies of gesture semantics focused precisely on this issue, modeled by way of formulas of SDRT (Segmented Discourse Represent Theory). We refer the reader to Lascarides and Stone 2009 for an SDRT treatment of gestures and their interaction with discourse structure.

1.4 Transcription conventions

For legibility, we use a non-standard font to transcribe gestures, in capital letters (capitals in normal font are standardly used to transcribe signs, i.e. sign language words, not gestures). We will use pictures whenever this can help evoke the intended gestures, sometimes preceded by a transcription for greater clarity. Pointing gestures are alphabetized from the dominant to the non-dominant side from the speaker's perspective. *IX-a* encodes pointing with a finger towards position *a* (which we will call a 'gestural locus').

[5] We write 'often lead to ungrammaticality' rather than 'always lead to ungrammaticality' because a pro-speech gesture may replace a modifier, and since modifiers can usually be omitted, this would be the case of the corresponding pro-speech gesture as well.

2 The Meaning Components of Pure Gestures

While it might be thought that pure gestures just contribute an unstructured informational content to the sentences they appear in, recent research argues for the opposite conclusion: their informational content is divided 'on the fly' among traditional slots of the inferential typology of language, which includes at-issue (= assertive) content, presuppositions, implicatures, and more.

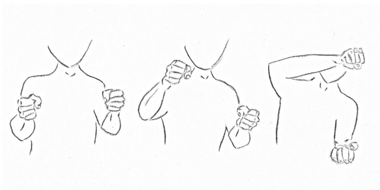
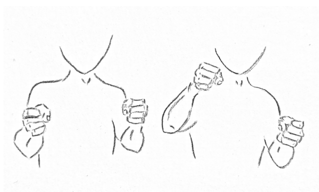
In some cases, this is expected. Diverse theories posit that an implicature arises as soon as an expression *S* competes with an expression *S*' which is more informative. A gesture could very well compete with another gesture, in which case one would expect gestural implicatures to arise. In other cases, the division of gestural content displays a productivity that is not expected by contemporary semantic theory. For instance, novel gestures can trigger presuppositions, despite the fact that presuppositions are usually thought to be lexically encoded. We will discuss these examples and a more subtle one, involving so-called 'homogeneity inferences'. We will then extend these results to iconic forms beyond gestures, which will further strengthen the case for the productivity of the division of gestural contents into different inferential components.

2.1 Implicatures in pure gestures

According to a variety of neo-Gricean theories (from Horn 1972 to Sauerland 2004, Spector 2006 and Chierchia et al. 2012), example (4)a yields the inference that not all group members attended because it competes with example (4)b, which is more informative.

(4) a. Some group members attended.
=> not all group members attended
b. All group members attended.

On virtually all recent theories, once appropriate alternatives are provided, implicatures follow by a productive algorithm. We thus expect that implicatures could be generated from entirely new and even non-standard material. This is indeed what was found for gestures (Schlenker 2019, Tieu et al. 2019). For instance, Tieu et al. 2019 showed with experimental means that example (5) gives rise to an inference that at the next mile marker John will turn the wheel, but not completely.

(5) *Context:* John is training to be a stunt driver. Yesterday, at the first mile marker, he was taught to TURN-WHEEL-COMPLETELY_  . . Today, at the next mile marker, he will TURN-WHEEL_  . .
=> at the next mile marker, John will turn the wheel, but not completely
(Tieu et al. 2019; illustrations: Marion Bonnet)

While the *not completely* inference could arise because the target gesture, glossed as *TURN-WHEEL*, just means 'turn exactly like this' (hence no more than is displayed by the gesture), a separate experimental condition (under negation) shows that this couldn't be all that happens. Rather, *TURN-WHEEL* has a general meaning of *turn the wheel* (obtained in particular under negation), and the *not completely* inference is obtained by competition with the broad wheel turning gesture, glossed as *TURN-WHEEL-COMPLETELY*.

The sentence in example (5) involves a direct implicature, one that is derived in a positive environment. Indirect implicatures can be obtained in negative environments, and they make the analysis more straightforward because the relevant inference couldn't be part of the literal meaning of the target gesture. As shown by Tieu et al. 2019, example (6) triggers the inference that at the next buoy John will turn the wheel (just not completely). This is plausibly derived by negating the stronger alternative, namely *he will not TURN-WHEEL*, hence John *will* turn the wheel (by elimination of the double negation).

(6) *Context:* John is training to be a stunt boat driver. Out by the first buoy, he decided to TURN-WHEEL-COMPLETELY, but at the second one he did not TURN-WHEEL. At the next buoy, he will not TURN-WHEEL-COMPLETELY.
=> at the next buoy, John will turn the wheel, but not completely
(Tieu et al. 2019)

Tieu et al. 2019 show in separate conditions that the gestures involved are unlikely to be mere codes for words because they have iconic implications that mere words would lack, for instance about the size of the wheel (for further arguments for productivity, see Sections 2.5 and 2.6).

In sum, as expected on virtually all contemporary theories, direct and indirect scalar implicatures can be triggered by gestures.

2.2 Presuppositions in pure gestures

Unlike implicatures, presuppositions were traditionally taken to be lexically encoded (e.g. Heim 1983). Numerous authors have argued that some or all are in fact triggered on pragmatic grounds (e.g. Grice 1981, Stalnaker 1974, Abbott 2000, Simons 2001, Abusch 2010, Schlenker 2010, Chemla 2010, Simons et al. 2010, Abrusán 2011, Romoli 2015, Tonhauser et al. 2013, Schlenker 2021b, Roberts and Simons 2024). Despite these efforts, there is no consensus on how presuppositions are triggered. Pro-speech gestures and visual animations have figured prominently in recent debates because they highlight the need for a non-lexical procedure. A key result is that diverse pro-speech gestures generate presuppositions, including when the gestures are unlikely to have been seen before.

Tieu et al. 2019 show with experimental means that two key presupposition tests, illustrated in example (7)a,b, suggest that the *TURN-WHEEL* gestural verb in example (8) triggers a presupposition that the agent is in the driver's seat.

(7) a. Did Ann stop / continue / regret smoking?
=> Ann smoked before
b. None of my students stopped / continued / regretted smoking.
=> Each of my students smoked before.

(8) a. *Context:* Jake and Lily are watching their four children ride bumper cars at the carnival. Each bumper car has two seats. As one of the bumper cars nears a bend in the track, the parents wonder:
Will Sally TURN-WHEEL?
=> Sally is in the driver's seat
b. *Context:* Blake and Diane are watching their group of friends ride bumper cars at the carnival. Each bumper car has two seats. As the various bumper cars near a bend in the track, they worry that:
None of their friends will TURN-WHEEL.
=> each of the relevant friends is in the driver's seat
(Tieu et al. 2019)

As illustrated in example (7)a,b, presuppositions project out of questions, and can give rise to universal positive inferences under *none* (further tests are discussed in Schlenker 2019 and Tieu et al. 2019). Precisely this behavior is displayed by the inference that the agent is in the driver's seat in example (8)a,b, which suggests that this inference is indeed a presupposition triggered by *TURN-WHEEL*. These results dovetail with the behavior of the gesture of unscrewing a bulb from the ceiling, discussed in (1): it too triggered a presupposition, namely that the bulb was positioned high.

In sum, gestures (including plausibly novel ones) can trigger presuppositions, which suggests that their informational content is productively divided among assertions and presuppositions.

2.3 Homogeneity inferences in pure gestures

Homogeneity inferences were investigated in speech in connection with the inferential behavior of definite plurals, illustrated in example (9) (e.g. Löbner 2000, Gajewski 2005, Spector 2013, Križ 2015, 2016, 2019, Križ and Spector 2021). In a context in which some presents were hidden for some children, example (9)a behaves roughly like example (9)a' in meaning that Mary will find all of her presents. However, the negation example (9)b doesn't mean that Mary won't find all of her presents, as would be expected if *her presents* meant roughly *all of her presents*; rather, it means that she won't find *any*. This is the initial puzzle raised by homogeneity inferences.

(9) a. Mary will find her presents.
=> Mary will find (almost) all of her presents
a'. Mary will find all of her presents.
=> Mary will find all of her presents
b. Mary won't find her presents.
=> Mary will find (almost) none of her presents
b'. Mary won't find all of her presents.
≠> Mary will find none of her presents

For Križ (e.g. 2015, 2016, 2019), the source of homogeneity lies in a lexical property of many (though not all) predicates, which 'want' their plural arguments to have all their parts in the predicate extension or all their parts outside of the predicate extension. If a predicate extension includes a plurality a and excludes a plurality a', it is undefined of a+a'; Križ shows that this property is responsible for the inferences in example (9).

Gestures (Schlenker 2019) and even visual animations (Tieu et al. 2019) can give rise to homogeneity inferences as well. All gestural (and even visual) examples discussed had the same structure: a gestural plural involving a repetition established an antecedent in a certain position; then a pointing gesture or a directed gesture (e.g. displaying an action of taking with both hands, directed to the established position) referred back to the antecedent, as illustrated in example (10). On Križ's theory, it is unsurprising that the English predicate *take* displays homogeneity effects. It is more interesting to note that the gestural predicate *TAKE-2-HANDED* displays the same property, highlighting the existence of homogeneity effects in the gestural domain.
Notation: Here $_{\text{left}}$ and $_{\text{right}}$ transcribe gestural loci, on the speaker's left and right respectively.

(10) *Context:* Sam is participating in a treasure hunt in the forest, and she is looking for crosses and coins. Very quickly, Sam will find [CROSS-rep3]$_{\text{left}}$ and [COIN-rep3]$_{\text{right}}$.
a. Sam will take IX-right / TAKE-2-HANDED-right.
=> Sam will take all of the coins
b. Sam will not take IX-right / TAKE-2-HANDED-right.
=> Sam will not take any of the coins
(Tieu et al. 2019)

In sum, homogeneity effects can be triggered by gestures, including by gestural verbs. Going forward, this should make it possible to create minimal contrasts between gestures that and gestures that don't trigger homogeneity effects, which in turn might help find the source of homogeneity effects.

2.4 Further inferential types

There are more inferential types than can be discussed here. For instance, supplements are usually defined as the meaning of appositive relative clauses. As a first approximation, they fail to interact with operators in whose scope they may be.[6] One characteristic behavior of supplements is that they usually yield deviance if they are forced to be interpreted in the immediate scope of a negative expression, as in example (11)b.

(11) a. One of these women helped her son, which saved him.
b. #None of these women helped her son, which saved him.

It was argued by some that gestures that follow the expressions they modify (i.e. post-speech gestures) can display the behavior of appositive relative clauses (Schlenker 2018, Esipova 2019; see also Ebert

[6] Systematic exceptions to the scopelessness of appositive relative clauses are discussed in Schlenker 2023a; to our knowledge, the scope of post-speech gestures has not been investigated in detail.

2024, who also presents an alternative in which post-speech gestures are gradiently at-issue). For instance, a lifting gesture (glossed as UP) is acceptable as a post-speech gesture under *one* but not under *none* in example (12)b, just as the close paraphrase with a *which*-appositive relative clause in example (12)a.

(12) a. One/#None of these 10 guys helped his son, which he did by lifting him.
a'. One/None of these 10 guys helped his son, which is surprising.
b. One/#None of these 10 guys helped his son – UP.

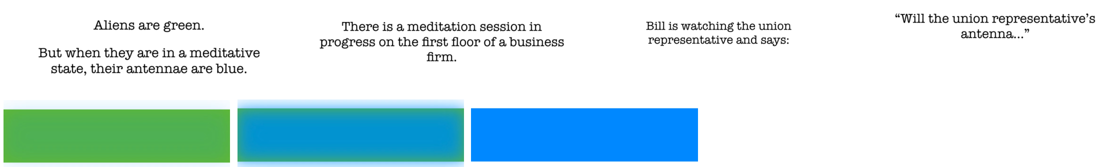
Similarly, it was argued that some gestures display the characteristic behavior of expressives, such as the term *Frog* to refer to French people (Schlenker 2019).

Supplements and expressives might differ a bit from our earlier examples in terms of productivity. The special behavior of supplements has sometimes been reduced to the presence of a 'comma intonation' (endowed with a complex lexical entry, Potts 2005), and if so it is unsurprising that it could apply to post-speech gestures just as it applies to appositive relative clauses.[7] As for expressives, the literature does not provide clear cases of novel gestures that behave like expressives, and thus the question of their productivity remains open.[8]

2.5 Inferential typology in iconic forms beyond gestures

Proving that some gestures are novel (i.e. have never been seen before) is a tall order. Tieu et al. 2019 make an argument for productivity by systematically replacing pure gestures with visual animations, and by showing that they too display the hallmarks of various inferential types: scalar implicatures, presuppositions, homogeneity inferences and supplements. Consider for instance presuppositions. The visual animation in example (13) represents an alien changing color from their normal state (green) to their meditating state (blue). For participants in the experiment, a change from green to blue triggered an inference that the aliens were not initially meditating. Crucially, this inference was obtained in a question, which made it plausible that this inference was a presupposition (see example (7)a above). This is in line with how natural language expressions and gestures behave: in change of state constructions, the initial state is presupposed (e.g. Abrusán 2011). Further tests and further pictorial representations involving changes of state were used to buttress the conclusion that visual animations too can trigger presuppositions.

(13) **Pictures from Tieu et al.'s videos testing presuppositions generated by visual animations**
(here: a change of state animation pertaining to an alien's antenna turning from green to blue; original video: https://youtu.be/U6dfs-XI2-4)



(green) (green+blue) blue

Going further, in view of the behavior of pure gestures and visual animations, we can expect that further visual forms could display similarly diverse inferential behaviors. This argument has been partly made about emojis: Tieu et al. 2024 show that certain emojis trigger supplements, while others trigger presuppositions.[9] The argument was further extended to the auditory domain. Thus Migotti and

[7] See Schlenker 2023b for the suggestion that post-speech gestures might be compared not just to appositive relative clause but also to gerunds.
[8] See, however, Schlenker 2019 for a speculation that the form of some gestural expressives (for instant referring to disabilities) might be modulated in iconic ways, which in turn might make it possible to create novel ones.
[9] A further claim is that, following ideas developed in Pierini 2021, some emojis have the semantic behavior of co-speech gestures, others of pro-speech gestures, and still others of post-speech gestures.

Guerrini 2023 show that the typology of linguistic inferences can be replicated with musical excerpts introduced within sentences and thus playing the role of 'musical gestures'.[10]

2.6 The connection to sign language classifier predicates

Since pure gestures are rare and sometimes marked, a perennial issue is whether they might be understood as codes for words, in which case the foregoing observations would only show that the words in question can trigger diverse inferential types—hardly a surprise. As mentioned, Tieu et al. 2019 include experimental conditions that show that experimental participants are sensitive to iconic implications of the target gestures, e.g. to the size or position of the wheel in the examples in examples (6) and (8). This makes it unlikely that participants *just* mentally replace the target gestures (or visual animations) with words. What is harder to exclude is that they reconstruct something with an iconic component, along the lines of: *turn the wheel like this*, where *this* makes reference to an iconic representation.

Interestingly, sign language research provides a complementary argument because iconic forms that *are* words can be gradiently modulated at will and yield the very inferential typology we have witnessed in gestures. The argument is made by the behavior of classifier predicates. As mentioned, they are words that have a lexically specified shape, but whose position and movement movement is free and is interpreted iconically (e.g. Emmorey and Herzig 2003, Zwitserlood 2012, Zucchi 2017). Crucially, classifier predicates were argued to trigger scalar implicatures, presuppositions and homogeneity inferences (Schlenker, 2021a; Schlenker et al., to appear). Since they are normal words of the language, it is highly implausible that they are mentally replaced with other words, which strengthens the view that iconic expressions can in themselves trigger diverse types of linguistic inferences.

2.7 Conclusions

While our survey is not exhaustive, it suggests that most or all of the inferential typology of language can be replicated with rare or novel gestures, and with further visual or auditory forms that couldn't have been seen in a linguistic environment before.[11]

One might just conclude from these findings that language is multimodal, and that gestures behave just like words and thus trigger the same inferential types. The multimodality of language, including when it comes to speech with gestures, is increasingly the default view in the field (e.g. Sandler and Lillo-Martin 2006, Goldin-Meadow and Brentari 2017). But this conclusion ('gestures are just part of spoken language') is insufficient, as it is compatible with a lexical account of the phenomena under discussion. The deeper theoretical challenge is to explain in great generality how diverse informational contents, including those of novel gestures and other forms one may not have encountered before, can be divided on the fly among established slots of the inferential typology of language.

3 The Functional Roles of Pure Gestures

We turn to cases in which pure gestures include functional (i.e. grammatical) elements. Sign languages have in this respect offered an important source of inspiration for gesture research. In several cases, historically unrelated sign languages tend to make use of the same forms to realize the same grammatical functions. Examples include pronouns realized as pointing, plurals and pluractionals realized by way of repetitions, telicity realized through sharp sign boundaries, and Role Shift, an operation sometimes analyzed as overt context shift, in which the signer shifts her body to adopt somebody else's perspective. In some cases, research on homesigners has found a similar convergence as well, for instance with pointing and repetition-based plurals (Coppola and So 2006; Coppola et al. 2013, Abner et al. 2015b). It is thus natural to ask whether the gestures of non-signers might display a

[10] Their original project included both onomatopoeias and musical excerpts, but somehow the experimental results were clearer with the latter.

[11] See also Schlenker 2021a and Fritzsche, to appear, for the observation that some pro-speech gestures can trigger conditionalized presuppositions (cosuppositions) that are otherwise thought to be characteristic of co-speech gestures.

similar convergence (for comparative studies of sign languages and gestures, see for instance Arik 2009 and Shaw 2013). Although the study could be developed with co-speech gestures, this would make it harder to delineate the respective role of spoken words and gestures. It is easier to focus on pure gestures (but see Colasanti 2023 for co-speech gestures that play a functional role). We will discuss in turn the grammatical functions of pointing, repetitions, gestural versions of Role Shift, and sharp gestural boundaries to mark telicity.

3.1 Gestural pointing

It is uncontroversial that co-speech pointing can have a deictic component, as when I say, faced with pictures of several employees, *I will promote HER [pointing]* (but see Fenlon et al. 2019 for differences between gestural and sign language pointing). This case can be replicated with pure gestures, fully replacing the pronoun with a pointing gesture: *I will hire IX* (where IX transcribes index pointing).

The example in (2) in the introduction went beyond deictic pointing because arbitrary positions were established for two antecedents, *a mathematician* and *a sociologist*. Depending on which position one then pointed to in the clause *I will hire IX*, one obtained the reading that the speaker will hire the mathematician or the sociologist. This mirrors the use of sign language loci to disambiguate anaphoric dependencies, in a case that is of special interest to theoretical linguistics: even though there is a clear anaphoric dependency between the pronominal expression and an existential quantifier (*a mathematician* or *a sociologist,* depending on the pointing), the pronoun is not in the syntactic scope (also called 'c-command domain') of the quantifier. These cases, sometimes called E-type or 'donkey' pronouns (because of examples involving donkeys in Geach 1962), have motivated the development of sophisticated semantic and/or syntactic mechanisms to explain why anaphora is still possible (Kamp 1981, Heim 1982; Heim 1990, Elbourne 2005). The sign language case has even been argued to provide evidence for some analyses over others, with the idea that loci can be the overt realization of logical variables (Schlenker 2011; see also Lillo-Martin and Klima 1990). In principle, the same arguments could be developed on the basis of pointing gestures (Schlenker 2020).

Once it is observed that arbitrary gestural loci can be associated with some antecedents, a further question arises: can such arbitrary loci be established for individuals or objects that are present in the discourse situation? In ASL (American Sign Language), this possibility is normally precluded: loci denoting speech act participants (and more broadly deictic elements) preferably correspond to their real position. The same finding can be replicated with gestures. In example (14)a, *Mary* is associated with a gestural locus on the right and *John* with a gestural locus on the left, and either one can be accessed by way of gestural pointing—in our example, pointing towards the right serves to denote Mary. The same anaphoric pattern is attempted in example (14)b, but *Mary* is replaced with *you*. The result is degraded, arguably because we have established an arbitrary gestural locus for an individua that is present in the discourse situation: one should point towards the interlocutor in this case.

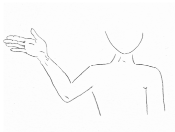
(14)
🖐-right 🖐-left
a. Tomorrow the boss will have a conversation with [Mary] and with [John]. And you know who the company will promote? ☞-right
🖐-right 🖐-left
b. ?? Tomorrow the boss will have a conversation with [you] and with [John]. And you know who the company will promote? ☞-right
(Schlenker 2020, 2022)

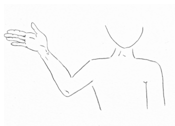
Along the same lines, Tieu et al. 2025 show experimentally that an arbitrary locus can be established for an indefinite antecedent, but not to refer to the addressee.

In sign language, besides index pointing, loci can be used to realize agreement-like modifications of certain verbs (called 'indicative' or 'agreement' verbs). An example involving object agreement appears in example (15): *TELL* must target the addressee when the object is second person, and some non-signer, non-addressee-related position when it is third person.

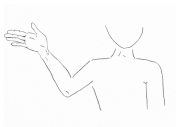
(15) I tell you I tell him/her[12]

Schlenker and Chemla 2018 argue (including with experimental means) that a similar distinction can be found in some gestural verbs. Thus in example (16)a, a slapping gesture targets a position on the speaker's right, and it is acceptable if the slappee is third person. But if the slappee is second person instead, as in example (16)b, the example becomes degraded: the slapping gesture must point towards the addressee.

(16) a. Your brother, I am going to  .

b. *You, I am going to  .
(Schlenker 2022; illustration: Marion Bonnet)

There is an additional twist to this story, however. As Schlenker and Chemla show experimentally, this condition is obviated under ellipsis, as illustrated in example (17)

(17) Your brother, I am going to  , and then you too.

The last clause means in essence *you too I am going to slap*. On standard views of ellipsis (e.g. Sag 1976), the missing VP after *too* is obtained by copying its antecedent in the first clause. But this should violate the condition that the gesture with a second person object target the addressee.

On closer consideration, however, this is exactly what one expects if the position targeted by the gesture behaves like a person feature. It is well-known that features inherited through agreement can be disregarded under ellipsis (e.g. von Stechow 2003), as is illustrated in example (18). The same behavior extends to person specifications of pointing sign in ASL (Schlenker 2014), and also to object agreement verbs in ASL.

(18) What is pronounced: Mary admires herself and you do too.
What is understood: Mary admires herself but you do too admire ~~her~~self.

In other words, the behavior of loci targeted by gestural verbs is consistent with a grammatical behavior found in signed and spoken languages alike. Whether this is the correct analysis remains to be determined, of course (see Esipova 2019 for relevant considerations).

3.2 Gestural plurals and pluractionals

In diverse sign languages as well as in homesigns, plurals of nominals can be realized in three ways. Often the noun may simply be unmarked, and be ambiguously singular or plural (as is the case in ASL and LSF [French Sign Language]). But plurality can also be realized by two different kinds of repetitions (Pfau and Steinbach 2006, Coppola et al. 2013, Abner et al. 2015b; see also Feldstein 2015).

[12] Bill Vicars, "TELL: The American Sign Language (ASL) Sign for 'Tell,'" ASL University, n.d., accessed November 20, 2021, https://www.lifeprint.com/asl101/pages-signs/t/tell.htm

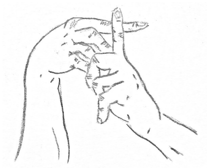
When repetitions are punctuated (clearly delineated and thus easy to count), the denotation is a plurality with the same number of objects as there are iterations. For instance, in ASL *TROPHY TROPHY TROPHY* stands for a set of three trophies. When repetitions are unpunctuated (without clear breaks between them) and thus hard to count, the meaning involves a vague threshold, and typically an at-least reading. So *TROPHY-rep3,* with three unpunctuated iterations, typically means 'at least three or four trophies'.

There are interesting twists. First, the arrangement of the repetition is usually iconically interpreted. For instance, if the iterations in *TROPHY*-rep3 are realized on a straight line, the plural will refer to a row of trophies. If they are realized as a triangle, it will refer to a triangular arrangement. Second, for unpunctuated repetitions, the number of iterations gives some idea of the threshold, with 5 iterations corresponding to a higher threshold than 3—although not necessarily to 5 vs. 3 objects, as the threshold is vague, and also context-sensitive.

Strikingly, there are related distinctions in gestures. Suppose one utters example (19), with different iterations of a gesture representing a cross (as in example (20)) in the position following *see*.

(19) *Context:* The addressee is taking part in a treasure hunt in churches. The speaker provides an indication about the location of the treasure.
If you enter a room and you see _________, you have reached the prize.
(Video: https://youtu.be/DXMsNtNZQ-U; Schlenker 2020, 2022)

(20) A gesture representing a cross



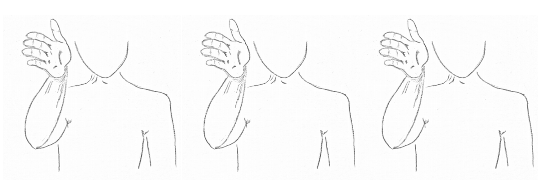
(illustration: Marion Bonnet)

Three horizontal punctuated repetitions of the cross gesture yield the meaning: *If you enter a room and see exactly three crosses arranged horizontally, you have reached the prize*. If the punctuated repetitions are arranged as a triangle, the condition is that the three crosses should form a triangle. Data are currently less clear with unpunctuated repetitions, but the threshold regarding the number of iterations is not precise. More work would be needed to establish whether a larger number of unpunctuated repetitions is associated with a higher threshold (Schlenker 2020).

In several sign languages, including ASL and LSF, just like repeated nouns can refer to pluralities of objects, repeated verbs can refer to pluralities of actions; they are thus 'pluractionals'. Sign language pluractionals come in two varieties: punctuated and unpunctuated, with 'exactly' meanings in the former case, and vague thresholds in the second (see Fischer 1973). Furthermore, just as for nominal plurals, verbal repetitions can have an iconic component. Thus if the verb *GIVE* is signed at an accelerating pace, one understands that the giving events also took place at this accelerating pace (Kuhn and Aristodemo 2017).

Related properties can be found in gestural verbs. Consider repetitions of a slapping gesture, as in example (21):

(21) (The slapping gesture is realized three times, separated by pauses.)

My opponent, I am going to  .
(Schlenker 2022; illustration: Marion Bonnet)

With three punctuated iterations, the meaning is that the speaker intends to slap his opponent three times. With unpunctuated repetitions, it becomes something like 'several times'. Iconic modification is possible: if the repetition starts slow and accelerates, it will probably be understood that the speaker intends to slap his opponent in this particular way.

In sum, although more work is needed on these matters, it seems that repetition-based plurals and pluractionals in pure gestures have some properties in common with sign language plurals and pluractionals.

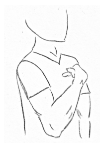
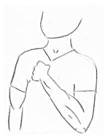
3.3 Gestural Role Shift?

Across sign languages, Role Shift that allows the signer adopt the perspective of another character. Role Shift is realized by several non-manual properties: (i) 'temporary interruption of eye contact with the actual interlocutor and direction change of eye gaze towards the reported interlocutor'; (ii) 'slight shift of the upper body in the direction of the locus associated with the author of the reported utterance'; (iii) 'change in head position'; (iv) 'facial expression associated to the reported agent' (Quer 2005, 2013). Role Shift comes in two varieties (e.g. Schlenker 2017a,b): Attitude Role Shift serves to report thoughts or words, with a quotational component; Action Role Shift describes an action in a particularly vivid way, often with an iconic component as well.

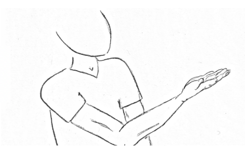
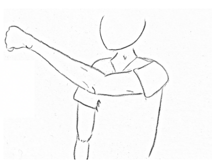
Some analysts (Quer 2005, 2013, Schlenker 2017a,b) take Role Shift to be an overt realization of 'context shift', a grammatical operation that shifts the context of evaluation of some or all indexicals. Others take Role Shift to incorporate a gestural, demonstrative component—with the idea that it signals that the words or actions were in relevant respects 'like' the words used by the signer (Davidson 2015).

Theory-neutrally, we can ask whether some pure gestures can be realized under a gestural version of Role Shift. In fact, there is a longstanding tradition that connects Role Shift to strategies of enactment in gestures (e.g. Liddell and Metzger 1998, Liddell 2003b, Lillo-Martin 2012, Quinto-Pozos and Parrill 2015; see also Emmorey et al. 2000). A gestural version of Attitude Role Shift is arguably exemplified in example (22):

Notation: $\mathbb{RS}_i$ indicates that the speaker shifts his body to adopt the position of a fictional character found in gestural locus i (here we will have $\mathbb{i} = \mathbb{a}$ or $\mathbb{i} = \mathbb{b}$). The gesture that follows $\mathbb{RS}_i$ is realized from this shifted position.

(22) I was standing next to -right [little Robin] and -left [little Francis], and I was holding a really yummy chocolate bar. And I asked: Who wants it? And so of course -right [little Robin] goes:  . And -left [little Francis] goes:  .
(Video: https://youtu.be/r0dhqgQk2k0, beginning; Schlenker 2022; illustration: Marion Bonnet)

A gestural version of Action Role Shift is arguably exemplified in example (23):

(23) Next thing I know, -right [little Robin] turns to -left [Francis] and  . And so -left [Francis]  .
(Video: https://youtu.be/r0dhqgQk2k0, end; Schlenker 2022; illustration: Marion Bonnet)

Tieu et al. 2025 show experimentally that, in gestural versions of Action and Attitude Role Shift alike, the operation is more acceptable when the speaker shifts to the position assigned to the character whose perspective is adopted than to another position. This too reflects standard constraints on Role Shift in sign language.

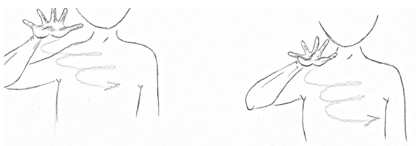
Thus sign language Role Shift might have a counterpart in gestures, although more detailed work will be needed to investigate its properties.

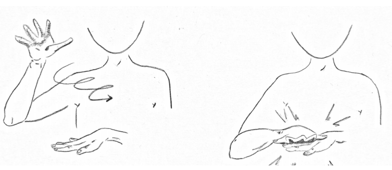
3.4 Telicity

Event descriptions are called *telic* if they hold of events that have a natural endpoint determined by that description, and they are called *atelic* otherwise. *Mary spotted John* and *Mary understood* have such a natural endpoint—the point at which Mary spotted John and came to an understanding, respectively; *Mary knew John* and *Mary reflected* lack such a natural endpoint and are thus atelic. Standardly (e.g. Rothstein 2004), a temporal modifier of the form *in* α *time* can modify telic VPs, whereas *for* α *time* modifies atelic VPs (e.g. *John reflected for a second* vs. *John understood in a second*).

Wilbur 2003 argued that the distinction between telic and atelic predicates is often realized overtly in ASL, namely by way of a "sharper ending movement to a stop, presumably reflecting the semantic end-state of the affected argument" (Wilbur and Malaia 2012). Strickland et al. 2015 revisit Wilbur's generalization, but in non-signers. They show that non-signers that have not been exposed to sign language still 'know' Wilbur's generalization about the overt marking of telic endpoints in sign language: when asked to choose among a telic and an atelic meaning (e.g. 'decide' vs. 'think') for a sign language verb they have never seen, they tend to be accurate in choosing the telic meaning in case endpoints are sharply marked. Furthermore, this result holds even when *neither* meaning offered to them is the actual meaning of the sign, which rules out the possibility that subjects use other iconic properties to zero in on the correct meaning.

Similar results arguably hold in pure gestures, as shown in example (24) (Schlenker 2020, 2022). In example (24)a, the gestural verb represents a parachute slowly rotating down, with the right hand, palm down, moving downwards without sharp gestural boundaries. The gestural verb in example (24)b is realized in a similar way, but the left hand is added to represent the ground, and there is a sharp boundary when the right hand hits the left hand, representing landing.

(24) a. When skydiving tomorrow, you will  ____ five minutes.

b. When skydiving tomorrow, you will  ____ five minutes.
(Related videos: https://youtu.be/bWraYBPSC-0; https://youtu.be/5xQN3aTxDXU; Schlenker 2022; illustrations: Marion Bonnet)

The contrasts are clear, and expected if Wilbur's generalization applies to gestural verbs: the empty position represented by ___ is preferably filled with 'for' for the gestural verb without a sharp ending (in example (24)a), and with *in* for the gestural verb with the sharp ending (in example (24)b). This is because an atelic meaning is obtained without a sharp ending, and a telic meaning with one. In sum, Wilbur's generalization appears to have a gestural counterpart.

3.5 Two directions: Universal Grammar or cognition

While further experimental studies would be needed to confirm all the hypotheses stated above, it is rather clear that certain specifications of gestures carry grammatical functions. The connection with sign language is particularly enlightening: if historically unrelated sign languages (and sometimes homesigns) converge on certain form-to-function mappings, it might not be unexpected that these can be found in gestures as well. Still, some of the gestural examples reviewed here are plausibly very rare, and seen for the first time by consultants or experimental participants. If so, they make a momentous point, namely that this convergence on certain form-to-function mappings is not purely due to functional pressure (transmission-related constraints over generations), but can arise on first exposure.

This could lead to two broad research directions. One possibility is that, in the visual modality, Universal Grammar does not just (innately) specify the abstract structure of some grammatical properties, but also certain form-to-function mappings—for instance, pronouns are realized by pointing, plurals by repetition, etc. An alternative direction is that some or all of these properties are based on non-language-specific properties. For instance, it was argued in Schlenker and Lamberton 2022 that the meaning of ASL repetition-based plurals can be derived from properties of precise and vague visual representations, combined with certain general mechanisms of pragmatic exploitation.

Tieu et al. 2025 address this question in a different way, by asking whether some properties of gestural grammar that are reminiscent of sign language grammar can be replicated outside the visual modality. For several phenomena involving loci, they provide a positive answer, and propose in particular that there can be 'sound loci'. This might be because loci have a cognitive source independent of the visual modality. However an alternative possibility is that loci are in fact bound to the visual modality, but can be generalized to the auditory domain. While the debate is open, understanding the roots of gestural grammar is an essential challenge for linguistics and cognitive science.

4 The Syntax of Pure Gestures

4.1 Goldin-Meadow's generalization

Goldin-Meadow et al. 2008 made a striking discovery: in a production task, speakers of syntactically very different languages (English, Mandarin, Spanish and Turkish) preferably used an SOV (= Actor Patient Action) word order when describing scenes in "silent gestures" (in our terms, pure gestures). Furthermore, this was independent of the word order of the experimental participants' native language (for instance, SOV gestural order was produced both by (SVO) English and (SOV) Turkish speakers). The authors further replicated their results in a non-communicative task in which participants "were asked to reconstruct the same events by using sets of transparent pictures". Here the order in which they stacked the transparencies also reflected an SOV order. Langus and Nespor 2010 further replicated these results in production and comprehension tasks involving Italian- and Turkish-speaking participants.

Later research added important qualifications to these findings. As summarized in Hall et al. 2013, Meir et al. 2010 found that speakers of Hebrew (an SVO language) and Turkish (an SOV language) used an SOV order in pantomimes when the subject was animate and the object was inanimate; this agreed with the findings of Goldin-Meadow et al. 2008. But when both participants were animate, and the action was thus 'reversible' (involving for instance a woman, a boy, and pushing), an SVO gestural order arose for both types of speakers. This finding was explained by the confusability of SOV when S and O are both animate and could thus equally have the roles of Actor and Patient. Gibson et al. 2013 developed an analysis of related results in terms of a rational 'noisy channel' theory of communication. Its main tenet is that speakers choose the best signal to communicate the intended meaning conditional on the assumption that some of the message might be corrupted. In reversible actions, upon the loss of an argument, SVO would still be partly interpretable (yielding SV or VO), but SOV wouldn't be—e.g. *boy push* wouldn't yield information as to whether the boy was doing the pushing or was being pushed. With non-reversible actions, involving for instance a boy, a box and some opening, the advantage of the SVO order disappeared (as one could infer from *boy open* that the boy was the agent and from *box open* that the box was the patient).

In a pantomime-based task with English-speaking participants, Hall et al. 2013 found a more nuanced picture: on the one hand, they confirmed the SOV preference for the representation of non-reversible events. On the other hand, they found that for the representation of reversible events, SOV was avoided and other orders emerged.

What all these studies have in common is the fact that in the representation of non-reversible events, an SOV order is preferred. But even in the non-reversible case, further distinctions are needed, as we will now see.

4.2 Intensional vs. Extensional Constructions

Several studies found that in silent gestures and in some sign language constructions, the choice between SOV and SVO in non-reversible constructions is determined by semantic considerations: extensional constructions give rise to an SOV preference, intensional constructions give rise to an SVO preference (a generalization we will correct in later subsections). Thus Schouwstra and de Swart 2014 argue that with extensional verbs, such as *throw*, an SOV order is preferred in silent gesture production, in line with Goldin-Meadow et al.'s (2008) results. But with intensional verbs such as *think of*, SVO is preferred. Importantly, all the scenarios were 'non-reversible' and thus correspond to a case which, according to earlier generalizations, should have yielded robust SOV preferences across the board.[13]

Schouwstra et al. 2019 confirm these generalizations (SOV for extensional objects, SVO for intensional ones), but from the perspective of interpretation. Having shown participants ambiguous gesture sequences, e.g. ones in which the gestural verb+object sequence could mean 'to build a house' (with an intensional object) or 'to climb a house' (with an extensional object), they found that SOV weakly favors the extensional reading while SVO weakly favors the intensional reading.[14]

Why should there be such a contrast? Schouwstra and de Swart 2014 argue that objects of intensional verbs are "more abstract and more dependent on the action" than those of extensional verbs, and that this motivates the word order facts. Christensen et al. 2016 propose that the target contrast follows from a principle of "structural iconicity" in which "the structure of events or relations between referents is replicated in the syntax of a spoken or signed utterance".

4.3 The Visibility Generalization

In a converging line of research, but based on Brazilian sign language (Libras) rather than on gestures, Napoli et al. 2017 found related generalizations and justified the import of the extensional vs. intensional distinction in terms of "visualization": unlike the object of an intensional verb, "the preexisting arguments of an extensional event are already somewhere in our mental picture before the predicate is articulated". While this intuition was intended to capture the distinction between extensional and intensional verbs, Schlenker et al. 2024 note that it leads one to expect a distinction among extensional verbs. Take the two sentences in example (25). Both involve extensional constructions.[15] But the ball would typically be visible before the swallowing in example (25)a, and after the spitting out in example (25)b.

(25) a. The crocodile swallowed a ball.
b. The crocodile spit out a ball (it had previously swallowed).

If Napoli et al.'s visualization-based analysis applies to sequences of pure gestures, one expects the gesture for the ball to appear before the gestural verb in sequences corresponding to example (25)a, and after the gestural verb in sequences corresponding to example (25)b.

On the basis of a small-scale study with consultants assessing the acceptability and meaning of French sentences involving sequences of pure gestures, Schlenker et al. 2024 argue that this expectation is met, as is stated in the generalization in example (26) (which dovetails with Napoli et al.'s as well as with Christensen et al.'s intuitions).

(26) **Visibility Generalization**
In classifier predicates and pro-speech gestures, the arguments are preferably realized

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[13] Schouwstra and de Swart follow Forbes (e.g. 2020) in taking three properties to characterize direct objects of intensional transitive verbs: "(1) resistance to substitution (i.e., Mary admires Mark Twain does not necessarily mean the same as Mary admires Samuel Clemens); (2) the possibility of a non-specific reading (such as in the sentence *Mary is looking for a man*, but not one in particular), or (3) existential neutrality (i.e., a sentence like *John is looking for a unicorn* is possible, in which the unicorn does not exist)."

[14] Schouwstra and de Swart's results subsume earlier findings by Langus and Nespor 2010 on gestural descriptions of speech- and thought-acts, and of Christensen et al. 2016 on manipulation events vs. construction events (a subclass of Schouwstra and de Swart's intensional category).

[15] For instance, using test (ii) in fn. 13, if the crocodile spit out a ball, there is a certain ball that the crocodile spit out.

a. before the verb if their denotations are typically visible before the denoted action, and
b. after the verb if their denotations are typically visible after the denoted action.

An example in which SVO order is regained appears in example (27). By contrast, with a minimally different scenario involving a gestural verb of swallowing, preverbal objects are preferred (it is not understood why an SOV preference is found in some studies and an SOV or OSV preference in others; correspondingly, we only seek to explain the preverbal vs. postverbal contrast).

(27) Hier en me promenant sur les bords du lac, je vois
Yesterday while me take-a-walk on the shores of-the lake, I see
CROCODILE **[un crocodile]** avaler BALL **[une balle]**.
CROCODILE ***[a crocodile]*** *swallow BALL* ***[a ball].***
Mais heureusement deux minutes plus tard,
But fortunately two minutes more late,
CROCODILE$_a$ ^ a-CROCODILE-MOVE-SPIT-b BALL$_b$.

(Video: https://youtu.be/mksxqTGuKEQ; Schlenker et al. 2024)

Schlenker et al. 2024 propose that the Visibility Generalization results from an iconic mode of composition. In line with their iconic semantics, the target gestures create a visual animation, and its component parts appear in the order they would in a comic or cartoon (this should yield interesting connections between these constructions and comics, see for instance Abusch 2011, 2020, Cohn 2013, Cohn and Schilperoord 2024). What is not fully explained is why objects of intensional verbs are conceptualized as appearing after rather than before the relevant speech- or thought-act. While Schlenker et al. 2024 give suggestive pictorial examples (from simplified comics) in which this preference might indeed be found, they leave open a full derivation of this fact.

4.4 The connection with sign language classifier predicates

The specific role of iconicity in driving non-standard word order is further highlighted by the fact that, within ASL, there is a contrast between standard signs, which have an underlying SVO order, and classifier predicates (the "super iconic" component of sign language), which often prefer preverbal objects (see for instance Liddell 2003a). Strikingly, however, ASL classifier predicates display a distinction between eat-up type predicates, which prefer preverbal objects, and spit-out-type predicates, which go with postverbal objects (Schlenker et al. 2024). Thus ASL classifiers obey the Visibility Generalization.

5 Conclusion

In sum, pure gestures have made three key contributions to recent theoretical linguistics. First, they show that most or all of the inferential typology of language is productive: the content of novel gestures is divided on the fly among established slots of this typology; this dovetails with results from visual animations and iconic constructions of sign language (classifier predicates), and highlights the question of the source of the inferential typology. Second, pure gestures show that certain grammatical components of sign languages, such as pronouns, plurals, Role Shift and telicity, might have some counterparts in the gestures of non-signers (in no way does this entail that sign languages are gestural systems, obviously). Here too, a key question pertains to the source of such regularities. Third, the existence of stable word order preferences in sequences of pure gestures raises a linguistic and cognitive puzzle. Recent findings suggest that the preference is driven by the existence of an iconic syntax that works on very different principles from standard syntax. Here too, findings from sign language classifier predicates strengthen this conclusion, and highlight the fruitfulness of a comparative approach to the visual modality.

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